

Data Collection and Missing Data

Announcements

- Chapter 8 and 18

BDA

Notation

- $y = (y_1, \dots, y_N)$, y is a matrix (y_i can be a p -vector).
- $I = (I_1, \dots, I_N)$ be a matrix of the same dimensions as y
- 'obs' = $\{(i, j) : I_{ij} = 1\}$ and 'mis' = $\{(i, j) : I_{ij} = 0\}$

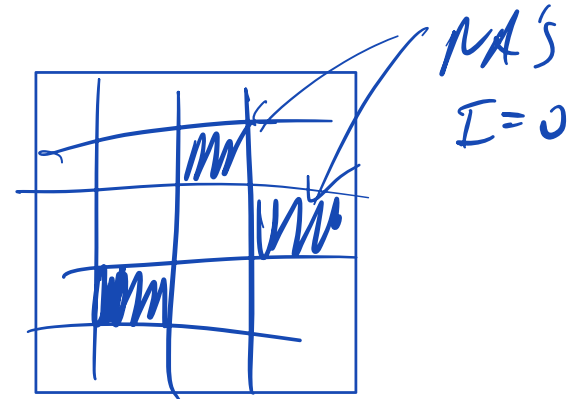
$y_{n \times p}$

$I_{n \times p}$

(indicators of
observability)

$y_{obs} = \{y_{ij} : I_{ij} = 1\}$

$y_{mis} = \{y_{ij} : I_{ij} = 0\}$



Assume: $P(Y, I | \Theta, \Phi, X) =$

$$\frac{P(Y | \Theta, X) P(I | Y, \Phi, X)}$$

(e.g.
Regression)

"complete data
model"



parameters
governing
observation
process.

Want: $P(\Theta | Y_{obs}, I)$

Compute posterior given
just observables.

observed data
likelihood

$$L(\Theta, \Phi) = P(Y_{obs}, I | \Theta, \Phi, X) =$$

$$\int \underbrace{P(Y, I | \Theta, \Phi, X)}_{\text{complete data likelihood}} dY_{mis}$$

complete data
likelihood.

$$P(\Theta, \Phi | Y_{obs}, X, I) \propto \underbrace{P(\Theta, \Phi | X)}_{\text{prior}} P(Y_{obs}, I | \Theta, \Phi, X)$$

$$P(\Theta | y_{\text{obs}}, X, I) = \int P(\Theta, \phi | y_{\text{obs}}, X, I) d\phi$$

observed data posterior.

Ignorability

(ignorability)

(no $I!$)

$$P(\Theta | y_{\text{obs}}, X, \underline{I}) = P(\Theta | y_{\text{obs}}, X)$$

$$\propto P(\Theta) P(y_{\text{obs}} | \Theta)$$

Special Case

You can "ignore"
the data observation
indicators.

Ignorability

- Two conditions imply ignorability:

- Missing at random (MAR)
- Distinct parameters

1. MAR $P(I | Y_{\text{complete}}, \phi, X) \stackrel{\text{MAR}}{=} P(I | Y_{\text{obs}}, \phi, X)$

2. Distinct parameters : $P(\theta, \phi | X) =$
 $P(\theta | X) P(\phi | X)$
"independent priors"

1+2: Ignorability.

Proof: $P(\Theta, \phi / y_{\text{obs}}, x, I) \propto$

$$P(\Theta, \phi / x) \int P(y / \Theta, x) P(I / \phi, y, x) dy_{\text{mis}} =$$

distinct params *MAR*

$$= P(\Theta / x) P(\phi / x) \int P(y / \Theta, x) P(I / \phi, y_{\text{obs}}, x) dy_{\text{mis}}$$

$$= \underbrace{P(\Theta / x) \int P(y / \Theta, x) dy_{\text{mis}}}_{P(\Theta / y_{\text{obs}}, x)} \underbrace{P(\phi / x) P(I / \phi, y_{\text{obs}}, x)}_{P(\phi / y_{\text{obs}}, x, I)}$$

Model Θ without
worrying about modeling
the observation
process (governed by ϕ)

Very Simple MAR example

$$y_i \overset{iid}{\sim} N(\theta, 1)$$

$$P(I_i = 1 \mid \phi, y, X) \overset{\text{assume}}{=} P(I_i = 1 \mid \phi, X)$$

"no y"

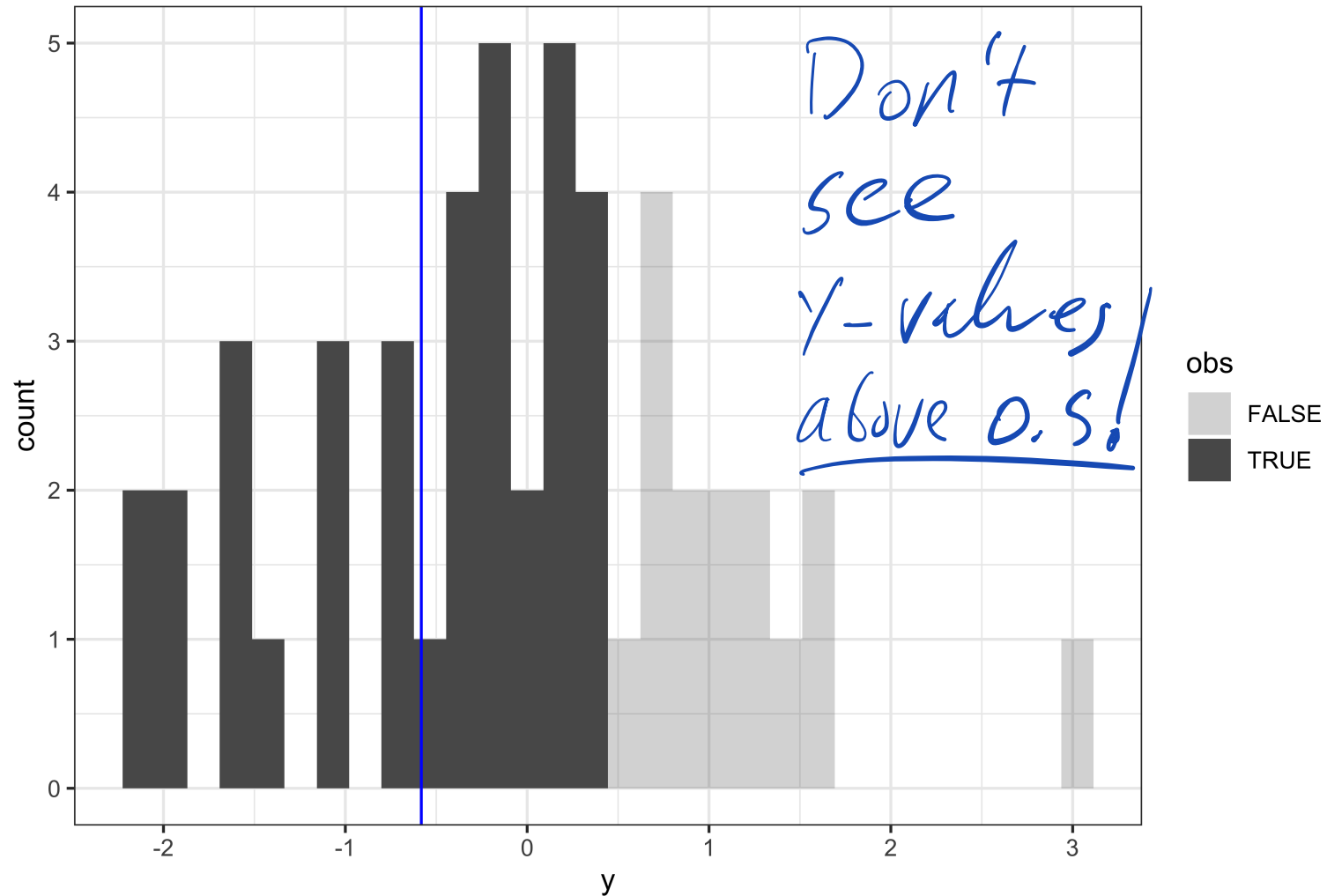
$$\begin{aligned} P(\theta \mid y_{obs}) &= \int \underbrace{\prod_{i=1}^{N_{obs}} N(y_i \mid \theta, 1)}_{\text{observed}} \prod_{i=N_{obs}+1}^{N_{obs}+N_{mis}} N(y_i \mid \theta, 1) dy_{mis} \\ &= \prod_{i=1}^{N_{obs}} N(y_i \mid \theta, 1) \underbrace{\int \dots}_{1} \end{aligned}$$

$$P(\theta) \propto \text{const}$$

$$\Rightarrow P(\theta \mid y_{obs}) = N(\bar{y}_{obs}, \frac{1}{N_{obs}})$$

Censoring and Truncation $y \sim N(\theta, 1)$

N=50, theta=0, phi=0.5



Censoring (known point)

$$y_i \stackrel{\text{iid}}{\sim} N(\theta, 1)$$

$$I_i = \begin{cases} 1 & \text{if } y_i \leq \phi = 0.5 \\ 0 & \text{if } y_i > \phi \end{cases}$$

$$P(I | y, \phi) \neq P(I | y_{\text{obs}}, \phi)$$

because I depends on possibly missing values! not MAR.

$$P(\theta | y_{obs}, I) \propto P(\theta) \int \underbrace{P(Y, I | \theta)}_{P(Y|\theta)P(I|Y, \theta)} dy_{mis}$$

$$\propto P(\theta) \int \prod_{i=1}^N N(y_i | \theta, 1) I[y_i \leq .5]^{I_i} I[y_i > .5]^{1-I_i} dy_{mis}$$

$$\propto P(\theta) \prod_{i=1}^{N_{obs}} N(y_i | \theta, 1) \int \prod_{i=1}^{N_{mis}} \underbrace{N(y_i | \theta, 1) I[y_i > .5]} dy_{mis}$$

$$\propto P(\theta) \prod_{i=1}^{N_{obs}} N(y_i | \theta, 1) \int_{.5}^{\infty} \prod_{i=1}^{N_{mis}} N(y_i | \theta, 1) dy_{mis}$$

$$\propto P(\theta) N(y_i | \theta, 1) \Phi(\theta - .5)^{N_{mis}}$$

$$\propto P(\theta) N(\theta | \bar{y}_{obs}, \frac{1}{N_{obs}}) \Phi(\theta - .5)^{N_{mis}}$$

Censoring (known point)

```
1 data {  
2   real ybar;  
3   int<lower=0> truncation_point;  
4   int<lower=0> ntotal;  
5   int<lower=0> nobs;  
6 } parameters {  
7   real theta;  
8 }  
9 model {  
10   target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) +  
11   (ntotal-nobs)*log(Phi(theta - truncation_point));  
12  
13 }
```

Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.
Chain 2 finished in 0.0 seconds.
Chain 3 finished in 0.0 seconds.
Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.
Mean chain execution time: 0.0 seconds.
Total execution time: 0.2 seconds.

A tibble: 2 × 10

	variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	lp__	-15.2	-14.9	0.703	0.319	-16.6	-14.7	1.00	1955.	2380.
2	theta	-0.184	-0.179	0.150	0.151	-0.430	0.0606	1.00	1470.	2229.

Censoring (unknown point)

Censoring (unknown point)

```
1 data {  
2   real ybar;  
3   real<lower=ybar> ymax;  
4   real<lower=0> ntotal;  
5   real<lower=0> nobs;  
6 } parameters {  
7   real theta;  
8   real<lower=ymax> phi;  
9 }  
10 model {  
11   target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) +  
12   (ntotal-nobs)*log(Phi(theta - phi));  
13 }
```

Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.

Chain 2 finished in 0.0 seconds.

Chain 3 finished in 0.0 seconds.

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.1 seconds.

Unknown censoring point

```
# A tibble: 3 × 10
  variable      mean  median      sd      mad      q5      q95  rhat ess_b...1 ess_t...2
  <chr>      <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>
1 lp__      -26.0    -25.7    1.07    0.785   -28.2   -25.0   1.00    1591.    1884.
2 theta     -0.0742  -0.0777  0.145   0.148   -0.310   0.172   1.00    2426.    2493.
3 phi        0.468    0.451  0.0548  0.0408    0.415   0.579   1.00    1708.    1292.
# ... with abbreviated variable names 1ess_bulk, 2ess_tail
```

Truncation (known point)

```
1 data {
2   real ybar;
3   real<lower=ybar> ymax;
4   real<lower=0> ntotal;
5   real<lower=0> nobs;
6   real<lower=ymax> truncation_point;
7 } parameters {
8   real theta;
9 }
10 model {
11   target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) -
12     nobs*log(1.0 - Phi(theta - truncation_point));
13 }
```

Truncation (known point)

Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.

Chain 2 finished in 0.0 seconds.

Chain 3 finished in 0.0 seconds.

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.2 seconds.

A tibble: 2 × 10

	variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	lp__	7.56	7.81	0.652	0.293	6.27	8.02	1.00	1775.	2194.
2	theta	-0.123	-0.123	0.225	0.228	-0.493	0.245	1.00	1485.	1625.

Truncation (unknown point)

Truncation (unknown point)

```
1 data {  
2   real ybar;  
3   real<lower=ybar> ymax;  
4   real<lower=0> ntotal;  
5   real<lower=0> nobs;  
6 } parameters {  
7   real theta;  
8   real<lower=ymax> truncation_point;  
9 }  
10 model {  
11   truncation_point ~ normal(0, 10);  
12   target += normal_lpdf(ybar | theta, 1.0/sqrt(nobs)) -  
13   nobs*log(1.0 - Phi(theta - truncation_point));  
14 }
```

Truncation (unknown point)

Running MCMC with 4 parallel chains...

Chain 1 finished in 0.0 seconds.

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Chain 3 finished in 0.0 seconds.

Chain 4 finished in 0.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.0 seconds.

Total execution time: 0.1 seconds.

A tibble: 3 × 10

	variable	mean	median	sd	mad	q5	q95	rhat	ess_b... ¹	ess_t... ²
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	lp__	4.53	4.89	1.11	0.802	2.19	5.60	1.00	1239.	1005.
2	theta	-0.124	-0.126	0.258	0.255	-0.536	0.299	1.00	1014.	879.
3	truncation_point	0.667	0.460	1.50	0.0493	0.416	0.743	1.00	1170.	750.

... with abbreviated variable names ¹ess_bulk, ²ess_tail

Multivariate Normal

MVN with missingness in STAN

Running MCMC with 4 parallel chains...

Chain 4 finished in 3.2 seconds.

Chain 2 finished in 3.4 seconds.

Chain 3 finished in 3.6 seconds.

Chain 1 finished in 3.8 seconds.

All 4 chains finished successfully.

Mean chain execution time: 3.5 seconds.

Total execution time: 3.9 seconds.

